Pokhara University

Everest Engineering College

Final Internal Assessment

SPRING-2025

Level: **Bachelor**

Program: **BE IT**

Faculty: **Science & Technology**

F.M. 100

P.M. 45

Time: 3hrs

Subject: Computer Network New (6th Semester)

Attempt all the questions.

1	a)	Define Computer Network. Explain about Network Model. And discuss the criteria necessary for an effective and efficient network?	2+5
	b)	Explain Protocol Stack, Interface and Services. Explain about protocol Hierarchies in detail?	4+4
2	a)	What do you mean by Transmission Media? Differentiate circuit switching and packet switching.	3+4
	b)	What is Forward Error Correction (FEC). What is the significance of hamming code in Error Detection, Explain with suitable Example.	8
3	a)	why access control of channel is essential? Describe in detail about IEEE standard 802 for LANs.	8
	b)	What are the design issues related to Transport layer? Explain TCP and its header segment.	7
4	a)	Discuss about Interior and Exterior Routing protocols. What is count to infinity problem in distance vector.	7
	b)	Pokhara university has 3 sub division located at Pokhara as head office, Kathmandu as Examination office and Biratnagar as contact office with 125,60 and 49 hosts respectively. Now you as network administrator design the network with below details. i) Calculate Broadcast address, Network address, usable address along with Subnet mask. ii)ISP provide IP address as 10.0.17.0/22	8

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5	a) What do you mean by congestion? Explain traffic shaping in details. b) What is the role of URLs and Web Browsers in WWW. Explain about DNS.	7 8
6	a) What is NMS. Explain about various functions of network management. b) What is the differences between substitution Ciphers and Transposition Ciphers? Give example of each type.	7 8
7	Write short notes on: (Any two) a) ATM b) NGN c) Piggybacking	2*5 =10

Best Wishes

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Spring- 2025

Level: Bachelor — **F.M.** 100
Program: BE IT(6th Semester) — **P.M.** 45
Faculty: Science & Technology — **Time:** 3hrs
Section: A/B
Subject: Data Science and Analytics

Attempt all the questions.

1 a) Why do we require data transformation? Explain by giving realistic examples. 7

b) You are given thirty numbers: 3.22, 5.38, 4.95, 4.87, 6.67, 4.46, 4.55, 5.42, 5.85, 4.81, 6.5, 5.66, 4.32, 5.21, 6.07, 3.97, 5.3, 5.37, 5.17, 5.35, 4.47, 5.98, 5.93, 5.5, 6.73, 4.33, 6.8, 4.76, 5.37, 6.15. Do you think the distribution is uniform, exponential, or normal? Justify using histogram plot. 8

2 a) Define a power law distribution and give real-life examples for it. Why power laws are called heavy-tailed distribution? Explain by comparing power laws to normal and exponential distribution? 8

b) Suppose you surveyed 5 male students and found their average daily spending on refreshments is Rs. 730, with a sample standard deviation of Rs. 200. A previous study shows that female students spend Rs. 810 per day on average. Using a t-test, determine if male students' daily spending is significantly different from that of female students. (Use $|t| > 1.8$ as the threshold for significance.) 7

3 a) How is the multivariate linear regression problem formulated in matrix notation? How is the solution obtained? Describe. 8

b) You are given the following data.

x	0.5	1.0	1.5	2.0
y	-2.3	0.5	9.3	20.8

Estimate the value of y at x = 1.3 using the Nadaraya-Watson regression estimator. Take bandwidth h = 0.2 7

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4 a) Explain Principal Component Analysis as a tool of identifying the latent variables in the data. 7

b) Suppose that while building a decision tree classifier, the current node corresponds to the following data.

A	B	Class label
T	F	Black
T	T	Black
T	T	Black
T	F	White
T	T	Black
F	F	White
F	F	White
F	F	White
T	T	White
T	F	White

Should the decision tree algorithm use the attribute A or the attribute *B* for splitting? Justify. You can use any purity index, Gini or entropy, for the problem.

5 a) Perform one iteration of k-means clustering given the following data. 7

Items	x	y
A	5	3
B	-1	1
C	1	-2
D	-3	-2

Take the number of centroids k=2. Assume that initially A and B are allocated to cluster 1 and C and D are allocated to cluster 2.

b) Why is logistic regression required? Describe how logistic regression models the probability $P(y = 1|\, x)$ where y is a binary variable. 8

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6 a) What is a stationary time series? Describe the ARMA model. 7

b) How can we be sure that an event A causes an event B? Differentiate correlation and causation with an appropriate example. 8

7 Write short notes on: (Any two) 2*5=10

a) Granger Causality

b) Statistical significance

c) The KDD process

Best Wishes

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